

Chapter 12

Embracing machine translation in L2 education: Bridging theory and practice in the AI Age

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This chapter examines the evolving role of machine translation (MT) in second language (L2) education. It first summarises recent research on MT's pedagogical applications, synthesising findings from systematic reviews that highlight MT's effectiveness in L2 writing when used appropriately. The chapter explores learner and teacher perceptions, noting a generally positive attitude among students but a more divided stance among educators. It presents a comprehensive framework for understanding factors influencing MT use in L2 writing, encompassing linguistic, personal, contextual, and ideological dimensions. The chapter introduces the concept of "MT as Augmented L2 Competence," offering a new paradigm for integrating MT into language learning. It discusses various models of MT instruction, including the Guided Use of MT (GUMT) model. Finally, the chapter proposes the Metacognitive Resource Use (MRU) framework, which positions learners as metacognitive agents capable of strategically utilising a wide range of language resources, including MT and generative AI (GenAI) tools. This integrated approach aims to foster autonomous learners who can effectively navigate the complex landscape of digital language learning resources.

1 Current status of MT use in L2 education

1.1 Pedagogical applications

As evidenced by the recent increase in publications on MT use for L2 learning and teaching, a wealth of research reports various pedagogical applications for MT in



Atsushi Mizumoto. 2026. Embracing machine translation in L2 education: Bridging theory and practice in the AI Age. In JC Penet, Joss Moorkens & Masaru Yamada (eds.), *Teaching translation in the age of generative AI: New paradigm, new learning?*, 233–248. Berlin: Language Science Press. DOI: 10.5281/zenodo.17641087 

L2 education. These applications include using MT as a resource for vocabulary acquisition (Lo 2025, 2024) and grammar practice (Lee & Kang 2024), fostering metalinguistic awareness through comparative analysis of original and machine-translated texts (Tsai 2019), and enhancing learner autonomy by promoting self-directed learning (Lee 2020).

Indeed, the proliferation of MT-related research over the past two decades has led to the emergence of several systematic reviews (Jolley & Maimone 2022, Klimova et al. 2022, Lee 2023, Ohashi 2024). These comprehensive analyses provide valuable insights into the current state of MT use in L2 education and highlight key trends and findings across multiple studies. Jolley & Maimone (2022) reviewed MT studies from 2000 to 2019, focusing on MT accuracy, educational applications, and perceptions of learners and teachers. They noted the improvement in MT quality over time and its effectiveness in L2 writing, while also highlighting uncertainties about its long-term impact on language learning. Klimova et al. (2022) analyzed 13 studies published between 2018 and 2021, specifically examining the impact of neural machine translation (NMT) in foreign language education. Their review found that NMT can be beneficial for language learning, particularly for advanced learners, but also identified a lack of teacher training in this area.

Lee (2023) conducted a systematic review of 87 studies from 2000 to 2019, investigating the effectiveness of MT in foreign language education. The review highlighted MT's usefulness in L2 writing, mixed perceptions among learners, and discrepancies between teacher and student perceptions. It also emphasised the need for further research on long-term learning effects and appropriate usage methods. Lee also conducted a meta-analysis with 12 studies that satisfied predefined criteria. It revealed that utilising MT had a positive impact on L2 learning with a small effect size ($g = 0.345$, 95% CI [0.201, 0.489]). The analysis of subcategories showed that using MT was effective in various linguistic areas, including lexical accuracy ($g = 0.616$, 95% CI [0.243, 0.676]), syntactic accuracy ($g = 0.562$, 95% CI [0.371, 0.754]), syntactic complexity ($g = 0.453$, 95% CI [0.181, 0.726]), orthography ($g = 0.741$, 95% CI [0.424, 1.059]), and overall writing quality ($g = 0.768$, 95% CI [0.547, 0.990]). However, the effect on lexical complexity was not statistically significant ($g = 0.253$, 95% CI [-0.058, 0.564]). These findings provide quantitative evidence for the positive impact of MT on various aspects of language learning, particularly in writing skills. This additional information strengthens the overall narrative by providing specific quantitative evidence of MT's effectiveness in various aspects of language learning. It supports the general consensus among the reviews that MT can be beneficial for L2 writing when

used appropriately, while also highlighting areas where its impact may be less pronounced (such as lexical complexity).

Ohashi (2024) examined 14 studies published between 2020 and 2022, reporting on the latest trends and key findings in MT research. This review emphasised that MT can be a good starting point for language learning, but its effectiveness varies depending on learner proficiency and target language. It also stressed the importance of communication between teachers and learners and the need for teacher training in MT use.

The systematic reviews collectively indicate that MT can be an effective tool for language learning when used appropriately. They emphasise that MT is particularly beneficial for L2 writing, helping students reduce lexico-grammatical errors and focus more on content (Klimova et al. 2022, Lee 2023). However, the reviews also point out that the effectiveness of MT use may vary depending on factors such as students' proficiency levels, target language pairs, and specific learning tasks (Ohashi 2024). Furthermore, these reviews underscore the importance of proper guidance and training for both students and teachers in effectively using MT for language learning. They highlight a gap between teachers' perceptions and students' actual practices, emphasising the need for better communication and mutual understanding (Jolley & Maimone 2022, Lee 2023). The reviews also call for more research on the long-term effects of MT use on language acquisition and the development of appropriate pedagogical strategies to integrate MT into language curricula effectively.

1.2 Learner and teacher perceptions

As reported above, MT use is widespread among language learners, who utilise these tools for a variety of purposes. Studies have found that a significant percentage of students, particularly those in higher education settings, report frequently using MT tools like Google Translate for language learning tasks, including writing and translation (Lee 2020, Tsai 2019). This widespread use persists despite learners often being aware of MT's limitations, particularly regarding accuracy and the potential for grammatical errors (Jolley & Maimone 2022).

Learners primarily view MT as a convenient and beneficial aid for language learning, using it for tasks such as looking up words and phrases, facilitating translation, and enhancing the revision process. While some learners express concerns about over-reliance on MT and its potential impact on learning, research indicates a generally positive attitude towards MT's role in supporting language acquisition (Chung & Ahn 2022).

In contrast to learner perspectives, the literature highlights a more divided stance among teachers regarding the use of MT in L2 education. While many teachers continue to express concerns about MT's potential to hinder language acquisition and facilitate academic dishonesty, others recognise its pedagogical potential as a valuable learning resource (Ducar & Schocket 2018).

The reluctance among teachers to embrace MT stems from several factors. Firstly, the historical association of MT with inaccurate translations, often referred to as a "bad model," and its perceived threat to traditional teaching methods contributes to skepticism (Lo 2025). Concerns about academic integrity, with MT being viewed as a form of cheating, also contribute to this reluctance. This often leads teachers to focus on detecting and preventing MT use rather than exploring its potential benefits (Jolley & Maimone 2022).

However, there is a growing movement among researchers and practitioners to integrate MT into language classrooms in a meaningful and responsible manner (Hellmich 2023, Lee 2020). This shift is driven by the understanding that MT use is now ubiquitous and cannot be effectively banned (Ducar & Schocket 2018). Instead, advocates of MT integration call for instructing learners about its strengths and weaknesses, equipping them to use it critically and ethically as a tool for language learning (Klimova et al. 2022).

1.3 Factors and views influencing MT use

The scoping review by Jiang et al. (2024), which synthesises 29 MT studies in L2 writing from 2009 to 2023, examines the factors influencing MT use in L2 writing, focusing on four main categories: linguistic, person/individual, contextual, and ideological factors, framed through cognitive, sociocultural, and critical theoretical perspectives.

- **Linguistic factors (cognitive perspective):** The review analyzes MT's role as a linguistic processor, exploring how the cognitive processing of MT impacts L2 writing. It explores the effects of different writing tasks on MT utilisation and assesses how the quality of input and output from MT systems affects the resulting text complexity and structure. These considerations are crucial in determining how both teachers and students perceive and employ MT, shaping its integration into educational practices.
- **Person/individual factors (sociocultural perspective):** MT is viewed as a mediational artifact/tool within social contexts, incorporating factors such as students' beliefs about MT's effectiveness, their writing strategies, L2

proficiency, and digital literacy. Additionally, teacher-related factors include their attitudes towards MT, pedagogical integration of MT, and ability to detect its use. These individual factors critically influence educational approaches and the effectiveness of MT.

- Contextual factors (sociocultural perspective): The role of MT is influenced by the dynamics within educational environments, such as peer interactions, the presence of MT training or scaffolding in instructional settings, and institutional policies that may either support or restrict MT usage. These contextual factors are pivotal in shaping the operational and situational environments for MT, impacting its perception and efficacy in educational settings.
- Ideological factors (critical perspective): Ideological considerations focus on language-related ideologies, including monolingual biases, standard language ideologies, and translanguaging stances, and how these influence the use and perception of MT. Such ideologies challenge conventional language education practices and influence how MT is integrated and impacts linguistic hierarchies within educational contexts.

Jiang et al. (2024) propose a layered conceptual framework to understand the factors influencing MT use in L2 writing. With ideological factors as the outermost layer, contextual factors encompass both person/individual and linguistic factors, with the latter being positioned at the innermost core. These layers interact to shape the core activities involved in the cognitive processing of linguistic factors. The model effectively illustrates the complex interplay from broad societal and ideological influences to specific individual and cognitive interactions, highlighting the necessity of considering a range of factors from societal ideologies to individual cognitive processes when integrating MT into L2 education.

Building on the comprehensive framework proposed by Jiang et al. (2024), we can see how ideological factors form the outermost layer influencing MT use in L2 education. This ideological dimension is particularly evident in two recent studies that highlight contrasting perspectives on MT use in academic settings.

Grieve et al. (2024) conducted a qualitative study exploring the ethical views of nursing and midwifery students on using AI machine translation software for university assignments. Their findings reveal a complex interplay of factors influencing students' ethical decision-making, including ownership of ideas, fairness and respect, and personal growth. Importantly, the study identified a tension between deficit-oriented and translanguaging perspectives among students, with

the latter referring to an approach that views learners' use of multiple languages as a resource rather than a problem.

The deficit-oriented perspective, often aligned with traditional language teaching methodologies, perceives L1 or MT use as detrimental to L2 acquisition. This perspective typically leads to policies discouraging or prohibiting MT use, viewing it as an impediment to authentic language learning. With this perspective, it is believed that that MT use may hinder the development of critical thinking skills in the target language and interfere with immersive language experiences.

Conversely, the translanguaging view adopts a more inclusive approach to language learning. This perspective values learners' entire linguistic repertoire as a resource (Wei 2018). Advocates of translanguaging argue that MT can be a valuable tool for accessing and leveraging learners' full range of linguistic knowledge, potentially enhancing both language awareness and learning outcomes. This view aligns with contemporary understandings of bilingualism and multilingualism, which conceptualise languages as part of an integrated communication system rather than as separate entities.

These contrasting perspectives on MT use in language education are further exemplified in the approaches educators and institutions take when addressing student use of MT. Jolley and Maimone's (2022) comprehensive review of three decades of MT research in language teaching and learning highlights two distinct approaches: the "MT as Cheating" approach, which leads to a Detect-React-Prevent Response, and the "MT as Resource" approach, which encourages an Integrate-Educate-Model strategy.

The "MT as Cheating" perspective, aligned with the deficit-oriented view, treats MT use as a form of academic dishonesty. This approach focuses on strategies to detect unauthorised MT use, react punitively, and prevent future occurrences. Proponents of this view recommend implementing clear syllabus policies against MT use, designing assignments resistant to MT use, and educating students about the pitfalls of relying on MT. This perspective often leads to policies that ban MT use outright, viewing it as incompatible with language learning goals.

In contrast, the "MT as Resource" approach, more closely aligned with the translanguaging view, sees MT as a potential tool for language learning. This perspective advocates integrating MT into the curriculum, educating students on its appropriate use, and modeling effective strategies for leveraging MT in language learning. Researchers like Stapleton & Kin (2019) and Niño (2020) argue for accepting the reality of MT use and finding ways to incorporate it meaningfully into language education. This approach acknowledges the ubiquity of MT in modern life and seeks to prepare students to use it critically and effectively.

The shift from the Detect-React-Prevent mindset to the Integrate-Educate-Model approach reflects a growing recognition of the inevitability of MT use in language learning contexts. As Ducar & Schocket (2018) note, the key question is no longer whether teachers can prevent learners from using MT, but rather how to help them use it ethically and effectively as part of their language learning journey.

These contrasting approaches to MT use in language education exemplify the broader ideological tensions identified in Grieve et al. (2024) and reflect the outermost layer of ideological factors in Jiang et al.'s (2024) framework. They demonstrate how deeply held beliefs about language acquisition and the role of technology can shape educational policies, pedagogical practices, and ultimately, students' engagement with and perceptions of MT in their L2 development process. This interplay between ideological stances and practical approaches underscores the complexity of integrating MT into language education and highlights the need for context-sensitive strategies that consider both the potential benefits and challenges of MT use in L2 learning and teaching.

2 Integrating MT into L2 education: A new paradigm

2.1 MT as augmented L2 competence

The translanguaging perspective and “MT as Resource” approach, implemented through the Integrate-Educate-Model strategy, provide a theoretical and practical foundation for incorporating MT into L2 education. Building upon these concepts, we can further conceptualise MT use in language learning through the lens of “MT as Augmented L2 Competence.” This model offers a visual representation of how MT can enhance learners' language abilities, particularly in bridging the gap between receptive and productive skills. By viewing MT as a tool for augmenting competence rather than replacing language learning, we align with the translanguaging idea of fluid language practices and the “MT as Resource” approach. The Integrate-Educate-Model strategy can then be applied to help learners effectively utilise MT to expand their augmented competence zone, while simultaneously developing their own language skills. This integrated perspective not only justifies the use of MT in language learning but also provides a framework for understanding its role in enhancing overall L2 proficiency.

The concept of MT as augmented L2 competence is illustrated in Figure 1, which provides a visual representation of how MT, and also GenAI such as ChatGPT, can enhance language learners' abilities.

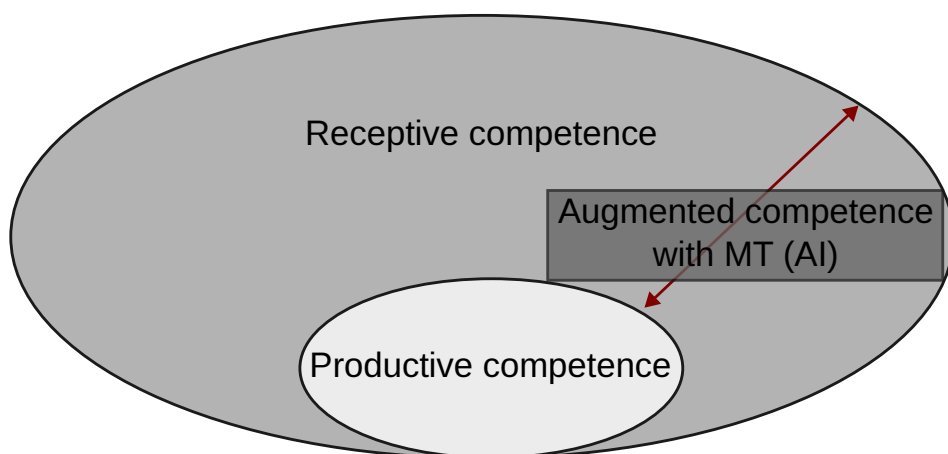


Figure 1: The concept of MT as augmented L2 competence

The figure demonstrates the relationship between receptive competence, productive competence, and the potential for augmented competence through MT use. Here is a breakdown of the key elements:

- **Receptive Competence:** This is represented by the larger, outer oval. It refers to the ability to understand the target language (L2), which is typically more developed than productive skills. For most L2 English language learners, their capacity to comprehend English exceeds their ability to produce it.
- **Productive Competence:** Shown as the smaller, inner oval, this represents the learner's ability to actively use the language. It is generally more limited than receptive competence, which aligns with theories like Swain's output hypothesis (1985), emphasising the importance of language production in second language acquisition.
- **Augmented Competence with MT (AI):** This is depicted by the dark gray area extending beyond the productive competence oval. It illustrates how MT can bridge the gap between what learners can recognise as correct (receptive knowledge) and what they can produce on their own.

Figure 1 suggests that MT can serve as a tool to augment learners' competence, particularly in areas where they can recognise correctness by sight but struggle to produce it accurately. This augmentation is especially beneficial for more proficient learners, as supported by previous studies (Klimova et al. 2022, Ohashi

2024). Higher proficiency learners tend to have a larger gap between their receptive and productive skills, providing more room for MT to assist in bridging this divide.

Importantly, this model underscores that there remains a strong rationale for studying English (or any L2). The augmented competence provided by MT is built upon the foundation of the learner's own language skills. Without developing one's own receptive and productive competencies, the benefits of MT augmentation would be limited. Furthermore, as learners' proficiency increases, they become better equipped to effectively utilise MT, maximising its potential as a learning tool.

This conceptualisation of MT as augmented L2 competence aligns with the findings from systematic reviews (Jolley & Maimone 2022, Lee 2023) that highlight MT's effectiveness when used appropriately, particularly for more advanced learners. It also supports the need for proper guidance and training in MT use, as the tool's effectiveness is contingent upon the learner's ability to critically evaluate and apply its output.

In sum, the model presented here provides a framework for understanding how MT can be integrated into language learning processes. It emphasises that MT is not a replacement for language study, but rather a tool that can enhance and extend learners' existing competencies, potentially accelerating their progress towards higher levels of language proficiency.

2.2 MT instruction for L2 learning

Niño (2009) proposed four models of MT use in L2 education: a "bad model", a "good model", vocational applications (particularly in translation-related fields), and as a computer-assisted language learning (CALL) tool. Initially, MT was employed as a "bad model", where students identified and corrected errors through post-editing, a process necessitated by the limited accuracy of early systems. In contrast, the "good model" involved using MT outputs as exemplars for students. These models, reflecting the evolution of MT technology and its pedagogical applications, illustrate a significant shift in focus. As MT technology has advanced, its primary use has transitioned to serving as a CALL tool, where it facilitates student engagement in solving language problems independently, as evidenced in recent studies (Lee 2020, Stapleton & Kin 2019, Tsai 2019).

Lee (2023) recommends that teachers should provide guidelines for using MT and explicitly teach effective strategies to students prior to using it, which leads to enhancing student performance, as supported by O'Neill (2016).

While many researchers discuss the need for explicit MT instruction in language classrooms, only a limited number of studies provide concrete examples (Cancino & Panes 2021, Chang et al. 2022, Mirzaeian 2021, O'Neill 2016, Ryu et al. 2022), all of which report positive results. Among these, Ryu et al. (2022) implemented a model called the Guided Use of MT (GUMT). The GUMT model activities were developed and implemented in an upper-elementary Korean as a foreign language course at a U.S. university. The GUMT model has five steps:

- **Instructional Session on MT Tools:** As Activity 1, Students review strengths and weaknesses of popular MT resources like Google Translate and Naver Papago, discussing differences in speech styles and politeness levels. As Activity 2, students engage in pragmatic and grammar evaluation exercises, assessing the appropriateness of translations by considering contexts such as formality and speaker-listener relationships.
- **Practice with Writing Assignments:** Throughout the semester, students apply learned concepts in five writing tasks, using MT to aid their translation from Korean (L2) to English (L1) and vice versa.
- **Reflection on MT Use:** After each writing assignment, students reflect on their use of MT, assessing its effectiveness and making adjustments based on their experiences.
- **Instructor Feedback:** Instructors provide written feedback on students' drafts, focusing on grammatical and pragmatic accuracy, and highlight areas needing correction.
- **In-Class Review Sessions:** Students refine their drafts in class based on instructors' feedback, discussing and revising highlighted issues to improve their understanding and application of MT in writing.

Ryu et al. (2022) reported that implementing the GUMT model significantly contributed to the development of MT use strategies and enhanced students' confidence and self-assessed fluency in L2 writing, as evidenced by analyses of both pre- and post-surveys and student reflections.

2.3 Learners as metacognitive agents

The role of learners as metacognitive agents in using MT and other language resources is crucial for effective language learning. Mizumoto (2023) proposed a

framework called Metacognitive Resource Use (MRU), which provides a comprehensive approach to understanding how learners can strategically utilise various language resources, including MT and GenAI tools like ChatGPT (Figure 2).

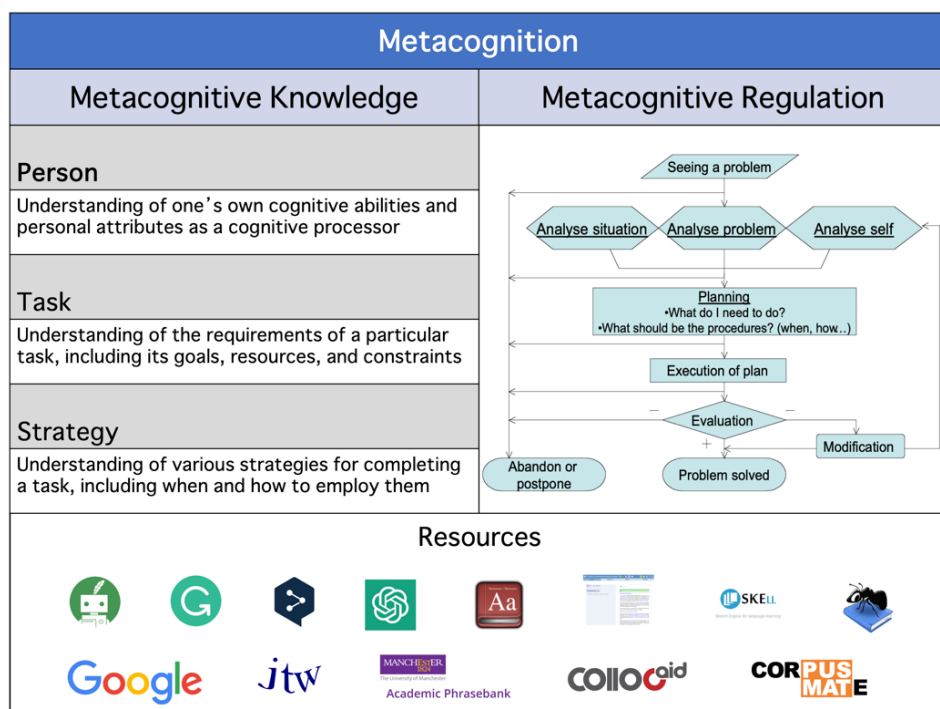


Figure 2: Framework of metacognitive resource use (adapted from Mizumoto 2023).

The MRU framework is grounded in metacognition theory and consists of two main components: metacognitive knowledge and metacognitive regulation. Metacognitive knowledge encompasses understanding of the person (self-knowledge), task (requirements and constraints), and strategy (available approaches). Metacognitive regulation involves the active management of this knowledge through planning, monitoring, and evaluating one's use of resources through employing learning strategies.

The framework visualises a range of resources that learners can utilise, including online dictionaries, corpus-based data-driven learning (DDL) tools like concordancers, web apps, search engines, GenAI systems like ChatGPT, MT tools, and grammar correction tools such as Grammarly. This reflects the real-world usage of tools by learners when engaging with language tasks.

For instruction on metacognitive resource use, Mizumoto (2023) suggests adopting principles from learning strategy instruction (Chamot & Harris 2019). This approach involves enhancing learners' metacognitive knowledge and imparting specific metacognitive and related cognitive strategies. The instructional model focuses on exposing students to a diverse array of tools and guiding them through the processes of planning, executing, evaluating, and adapting their strategic resource use.

It is important to note that the GUMT model employed by Ryu et al. (2022) aligns well with this approach. The GUMT model does not rely on a single tool but incorporates various resources, including different MT platforms and reflective practices. Furthermore, the instruction model in Ryu et al. (2022) shares similarities with both DDL and strategy instruction approaches, demonstrating that these methodologies can be discussed within the same theoretical framework that Mizumoto (2023) advocates for integrating DDL and GenAI.

This integrated perspective allows for a more comprehensive approach to language resource use. Instead of focusing solely on MT, learners should be encouraged to become metacognitive agents capable of effectively utilising a wide range of resources, including GenAI tools like ChatGPT. By adopting this approach, we can foster the development of autonomous learners who are adept at selecting and using the most appropriate tools for their language learning tasks.

The MRU framework not only enhances learners' awareness and use of MT but also extends to their conscious utilisation of GenAI tools. This holistic approach has the potential to cultivate truly autonomous learners who can navigate the complex landscape of language learning resources effectively.

Moreover, the research findings accumulated in the field of MT should not be disregarded but rather re-examined and validated within this integrated and extended framework. By doing so, we can ensure that the valuable insights gained from MT research continue to inform and enhance our understanding of language learning processes in the era of diverse digital resources.

In conclusion, by positioning learners as metacognitive agents within the MRU framework, we can create a more comprehensive and effective approach to language learning that embraces the full spectrum of available resources, from traditional tools to cutting-edge AI technologies.

2.4 Distinguishing MT from GenAI

While this chapter has primarily focused on MT, it is important to acknowledge the emerging role of GenAI tools like ChatGPT in L2 education. Although both MT and GenAI can assist with language tasks, they differ fundamentally

in their capabilities and potential applications. Traditional MT systems are designed specifically for translation between languages, focusing on maintaining semantic equivalence while adapting to target language conventions. In contrast, GenAI systems can engage in open-ended language generation, including not only translation but also extending to explanation, summarisation, paraphrasing, and interactive dialogue.

The differences between these technologies extend to their ability to handle context and maintain coherent interactions. While MT primarily processes text at the sentence or paragraph level, GenAI can maintain longer conversations, take broader context into account, and engage with complex prompts across multiple exchanges. This enhanced contextual awareness allows GenAI to serve not just as a tool for language conversion, but as an interactive partner in the language learning process.

These fundamental differences lead to distinct educational applications. MT serves primarily as a translation tool, though it can be used pedagogically for language comparison and error analysis. GenAI, however, can potentially function as an interactive tutor, providing explanations, examples, practice exercises, and feedback on language use. This broader functionality suggests that while MT typically serves as a standalone tool for translation tasks, GenAI has the potential to integrate and potentially replace multiple language learning tools, including dictionaries, grammar checkers, and corpus resources.

3 Concluding remarks

Looking ahead, the integration of MT, GenAI, and other AI-powered language tools in L2 education is likely to continue evolving rapidly. Future research should focus on developing and empirically testing comprehensive frameworks like the MRU, which embrace a wide range of digital resources beyond just MT. As the capabilities of GenAI continue to expand, there is a particular need to understand how these tools can complement or potentially transform traditional MT use in language learning contexts. There is a need for longitudinal studies to assess the long-term impact of these tools on language acquisition and learner autonomy, specifically examining how the interactive capabilities of GenAI might differ from traditional MT in supporting language development. Additionally, as AI language models become more sophisticated, practitioners and researchers will need to continually adapt their approaches, balancing the benefits of these tools with the core objectives of language learning. The field may see a shift towards more personalised, AI-assisted language learning experiences, necessitating new pedagogical strategies and ethical guidelines for their implementation

in L2 learning and teaching. Understanding the distinct roles and potential synergies between MT and GenAI will be crucial for developing effective educational practices in this rapidly transforming field.

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